

Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language
Models for Histopathology Image Analysis

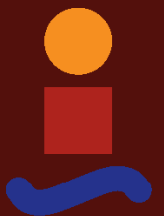
Author: Leandro Candau Sánchez de Ybargüen

Tutors: Miguel Ángel Gutiérrez Naranjo

Miguel Cárdenas Montes

Dpto. de Ciencias de la Computación e IA
Escuela Técnica Superior de Ingeniería Informática
Universidad de Sevilla

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Author:

Leandro Candau Sánchez de Ybargüen

Tutors:

Miguel Ángel Gutiérrez Naranjo

Profesor Titular

Miguel Cárdenas Montes

Co-director

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Image Analysis

Autor: Leandro Candau Sánchez de Ybargüen
Tutor: Miguel Ángel Gutiérrez Naranjo
Tutor: Miguel Cárdenas Montes

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*Leandro Candau Sánchez de Ybargüen
Sevilla, 2026*

Resumen

- Contexto: clasificación de imágenes histopatológicas con datos etiquetados limitados y etiquetas repetidas.
- Enfoque: aprendizaje contrastivo imagen-texto al estilo CLIP con codificadores fundacionales congelados.
- Datos: Kaggle LC25000, 5 clases de tejido de pulmón y colon.
- Diseño experimental: línea base controlada + matriz de ablaciones (una variable a la vez).
- Resultados (parciales): línea base 63.6 % accuracy / 62.7 % macro-F1; PLIP cero-disparo +7.3 macro-F1; ajuste fino parcial (últimas 30 capas) +27.1 macro-F1 (89.8 % / 89.8 %).

Abstract

- Context: histopathology image classification under limited labelled data and repeated labels.
- Approach: CLIP-style image-text contrastive learning with frozen foundation-model encoders.
- Data: Kaggle LC25000, five lung and colon tissue classes.
- Experimental design: a controlled baseline + a single-variable ablation matrix.
- Results (partial): baseline 63.6 % accuracy / 62.7 % macro-F1; PLIP zero-shot +7.3 macro-F1; partial fine-tuning (last 30 layers) +27.1 macro-F1 (89.8 % / 89.8 %).

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1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data - recurring limitation in this domain.
- Foundation models - able to do different tasks with little task-specific supervision.
- Project question: "How does CLIP transfer to histopathology with different backbones? Where are its limits?"
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction -> semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot -> image-level classification, same domain, reframed question.
- Testbed: LC25000. Reference point: PLIP.

1.3 Research Questions

- RQ1: Frozen general-purpose CLIP transfer to LC25000 — quality and failure modes.
- RQ2: Pathology pretraining contribution (PLIP vs the general-purpose baseline).
- RQ3: Feature-adaptation contribution (partial unfreezing of the image encoder).
- RQ4: CLIP-paradigm contribution vs a plain image classifier on the same frozen features.
- RQ5: Stain normalization → shortcut-learning probe (Macenko / Reinhard / Vahadane head-to-head).
- RQ6: Biomedical text encoder (BioBERT / PubMedBERT) under short prompts.
- RQ7: Out-of-distribution transfer (external colon dataset).

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same controlled baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment in Chapter 4.

Measured claims:

- **Pathology-domain pretraining.** PLIP zero-shot vs the general-purpose baseline: +7.3 macro-F1 (69.8 % / 70.0 % vs 63.6 % / 62.7 %), with no LC25000 training. Gain concentrates on benign tissues; the two cancer subtypes (lung_aca, colon_aca) tie.
- **Feature-level adaptation.** Last 30 layers of ResNet50 unfrozen, LR dropped $10\times$ to 10^{-4} , everything else held constant: +27.1 macro-F1 (89.8 % / 89.8 %). Broad-based; every class $F_1 \geq 0.85$; lung_aca moves 0.48 $\rightarrow$ 0.85.
- **CLIP architecture vs plain CE classifier on the same features.** A 2×2 comparison at each adaptation budget. Frozen regime: tied within noise (+1.0 macro-F1 to plain CE, 0.6369 vs 0.6273). Unfreeze30 regime: CLIP wins +2.6 macro-F1 (0.8981 vs 0.8723), broad-based across classes. The CLIP architecture neither costs nor benefits accuracy at the frozen-features ceiling but gains a small consistent advantage when features adapt, plausibly via implicit regularisation from the InfoNCE loss.

Methodology:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Artifact bundle (fixed-seed split, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits constraining how far each delta can be generalised.

Pending ablations, scoped and implemented, to be folded in as measured claims once the corresponding run lands:

- Stain normalization head-to-head: Macenko, Reinhard, Vahadane vs raw inputs.
- Biomedical text encoder swap: BioBERT and PubMedBERT vs frozen BERT base.
- Out-of-distribution transfer of the baseline checkpoint to an external colon dataset.

1.5 Document Structure

- Chapter map to be added in the prose pass.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

BOMMASANI ET AL., 2021

- Foundation models framing (Bommasani 2021).
- CLIP and multimodal contrastive learning (Radford 2021) — the paradigm under study.
- Self-supervised visual pretraining for medical imaging — DINO, DINOv2.
- Pathology shift: per-tile supervised CNNs → foundation models (2016–2024).
- Pathology-specific foundation models:
 - TransPath (2022) — transformer + SSL.
 - Phikon (2023) — masked image modelling.
 - PLIP (2023) — CLIP on medical Twitter.
 - CONCH (2024) — larger pathology VLM.
 - BiomedCLIP, MUSK — recent entrants.
- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko, Reinhard, Vahadane).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
 - Open-vocabulary classification (new classes via prompts proven by adding new samples).
 - Semantic similarity output (cosine vs natural language).
 - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder + shared embedding space.
- InfoNCE loss, learnable temperature.
- In-batch false-negative regime at small batch + few classes.
- Prompt-based zero-shot classification.
- CLIP-paper sample-level retrieval → doesn't apply (class-shared prompts).

3.4 Image Encoders

- ResNet (CNN baseline).
- Vision Transformer (ViT).
- DINO / DINOv2 (self-supervised).

Two independent encoders project image and text into a shared d -dimensional embedding space. The cosine-similarity matrix across a batch is the input to the symmetric InfoNCE loss.

Figure 3.1 CLIP dual-encoder architecture used in this project..

3.5 Text Encoders

- Transformer + BERT — text encoder used by baseline.
- BioBERT / PubMedBERT — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP and a plain image classifier

- PLIP zero-shot (Huang 2023) — pathology-pretrained CLIP, frozen, no LC25000 training.
- Plain image classifier — Dense(5)+softmax on frozen ResNet50 trained with cross-entropy; no text encoder, no CLIP architecture.
- Together: PLIP isolates the pretraining-domain contribution; the plain classifier isolates the CLIP-architecture contribution.

3.7 Dataset: Kaggle LC25000

- Borkowski 2019.
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

Table 3.1 LC25000 class counts and stratified split sizes..

Class	Total	Train	Val	Test
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

Five-class sample: "benign lung", "lung adenocarcinoma", "lung squamous cell carcinoma", "benign colon", "colon adenocarcinoma".

Figure 3.2 Example image per class from LC25000..

3.8 Preprocessing and Input Pipeline

- Image: resize 224×224 , ResNet-style preprocessing, no further augmentation.
- Text: BERT WordPiece, fixed sequence length.
- Split: stratified 80/10/10, seed 42, persisted.

3.9 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) $\rightarrow$ 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

Table 3.2 Baseline hyperparameters. Ablations change exactly one row..

Hyperparameter	Value
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.10 Evaluation Protocol

- Classification: `argmax cosine(image, class-prompt)`.
- Global: accuracy, balanced accuracy, macro-F1, weighted-F1.
- Per-class: precision, recall, F1.
- Confusion matrices: raw + row-normalized.
- Embeddings: shared image+prompt UMAP, centroid-to-prompt heatmap.
- Class-level retrieval, with explicit positive definition.

3.11 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in §4.1.2.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH / BiomedCLIP / MUSK. Deferred to future work.

3.12 Computational Setup and Reproducibility

- Hardware: Google Colab T4, local fallback.
- Stack: TensorFlow + KerasHub; PyTorch + transformers for PLIP only.
- Pinned library versions.
- Seeds + PYTHONHASHSEED + `tf.config.experimental.enable_op_determinism()`.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research questions

- **RQ1** — What baseline classification and generalization behavior does a frozen general-purpose CLIP pipeline achieve on LC25000, and what failure modes appear?
- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learn one task, generalize to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalization, an input-side preprocessing that targets a known shortcut in histopathology, help when we evaluate on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT, PubMedBERT), instead of generic BERT, improve performance under short class-name prompts?
- **RQ5** — Does the image encoder’s *pretraining choice* matter? Specifically: (i) replacing the general-purpose ResNet50 with a Visual Transformer pretrained on histopathology image–text pairs (PLIP/OpenPath), and (ii) replacing it with a Visual Transformer pretrained self-supervised on general natural images (DINOv2), as a control that isolates “ViT architecture and self-supervision” from “histopathology pretraining specifically”.
- **RQ6** — How well does each variant generalize to data the model has never seen, evaluated against three external datasets across two organs that span shift severity (close-shift colon, different-hospital colon, different-organ lung)?

4.1.2 Experiment matrix

- Each row below is one experiment. Each one fixes the rest of the pipeline and changes exactly one axis relative to the baseline.
- All variants are evaluated on the same training-source test split (LC25000) and the same three external datasets (Section 4.1.3).

4.1.3 Training source and external datasets

- **Training source:** LC25000 (5 classes, 80/10/10 stratified split, dedupe applied; see Methodology). Macro-F1 on the LC25000 test split is what we call “training source results” below; for the rigor-minded reader this is the in-distribution evaluation.
- **External datasets** (data the model has never seen during training; we call this “generalization to external datasets” below; for the rigor-minded reader these are out-of-distribution evaluations):
 - **NCT-CRC-HE-7K** (close-shift colon, 1 974 patches; mapped 2-class: NORM ↔ benign colon, TUM ↔ colon adenocarcinoma).

Table 4.1 Experiment matrix. Each experiment isolates exactly one axis of variation relative to the baseline. Sections §4.2–§4.7 follow this order, ending with image-encoder pretraining (the chapter’s headline finding)..

Experiment	Image encoder	Text encoder	Loss	Preprocessing	Axis tested
Baseline (§4.2)	ResNet50 (ImageNet)	BERT base	InfoNCE	RGB / 255	reference
Stain normalization (§4.3)	ResNet50 (ImageNet)	BERT base	InfoNCE	Macenko / Reinhard / Vahadane	input-side
PLIP zero-shot (§4.4)	ViT-B/32 (PLIP / OpenPath)	PLIP text encoder	none (zero-shot)	CLIP mean/std	reference: no LC25000 supervision
Foundational vs plain (§4.5)	ResNet50 (ImageNet)	— (no text)	cross-entropy	RGB / 255	architecture: CLIP paradigm vs plain classifier
Text encoder (§4.6)	ResNet50 (ImageNet)	BioBERT / PubMedBERT	InfoNCE	RGB / 255	text encoder pretraining
Image encoder (§4.7)	ViT-B/32 (PLIP); ViT-B/14 (DINOv2)	BERT base	InfoNCE	varies	image encoder pretraining

- **Chaoyang** (different-hospital colon, 4060 patches scored from a 6 160-image release; mapped 2-class: normal ↔ benign colon, adenocarcinoma ↔ colon adeno).
- **LungHist700** (different-organ lung, 359 images at 20× from a ~ 691-image release; mapped 3-class: normal/AdC/SqC ↔ benign lung / lung adeno / lung squamous).
- **Restricted-argmax evaluation:** each variant exposes all 5 LC25000 class prompts at inference; we argmax only over the LC25000 classes that map to the external dataset’s organ (2 colon classes for NCT/Chaoyang, 3 lung classes for LungHist700). Misclassifications into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis.
- **Pure inference:** every variant reuses its LC25000-trained checkpoint. No fine-tuning on the external sets.
- **Per-dataset reporting:** results are presented per dataset, not averaged. Aggregating across datasets would hide the cross-organ regime change surfaced in Section 4.7.

4.1.4 Metrics and visualizations

- **Macro-F1** is the headline metric (class-balanced; reflects per-class precision/recall tradeoffs better than raw accuracy on uneven class supports).
- Accuracy, balanced accuracy, weighted F1 and per-class precision/recall/F1 reported alongside for completeness.
- **Confusion matrices** (raw and row-normalised) per variant per dataset — the most direct view of how a model distributes errors across classes.
- **Training-source UMAP:** image embeddings + class-prompt embeddings projected into the same 2-D fit (cosine metric). Reads: cluster separation, cluster alignment to prompt markers.
- **Cross-distribution UMAP grid** (per OOD dataset, per variant): a 2×3 grid showing LC25000 dots overlaid with external-dataset triangles in one fit. Reads: does the external dataset co-locate with the matching LC25000 class cluster, or sit in its own region of feature space?

4.2 Baseline: frozen ResNet50 and frozen BERT

4.2.1 Configuration

- Frozen ResNet50 image encoder (general-purpose, ImageNet-pretrained).
- Frozen BERT base (uncased) text encoder.
- Trainable projection heads (256-d shared embedding space) and learnable temperature τ .
- Symmetric InfoNCE loss, AdamW optimiser ($\text{lr} = 10^{-3}$, $\text{wd} = 10^{-4}$).

- Max 30 epochs, EarlyStopping (patience=5, restore best weights), seed 42.
- Composed prompt format: “A histopathology image of {class_name}.”.
- Stratified 80/10/10 split.

4.2.2 Training behaviour

- Training and validation InfoNCE loss converge together; the gap is small, no obvious overfitting visible at the converged epoch.
- Learnable temperature τ drifts from its CLIP initialisation of 0.07 toward a smaller converged value, sharpening the cosine-similarity distribution.

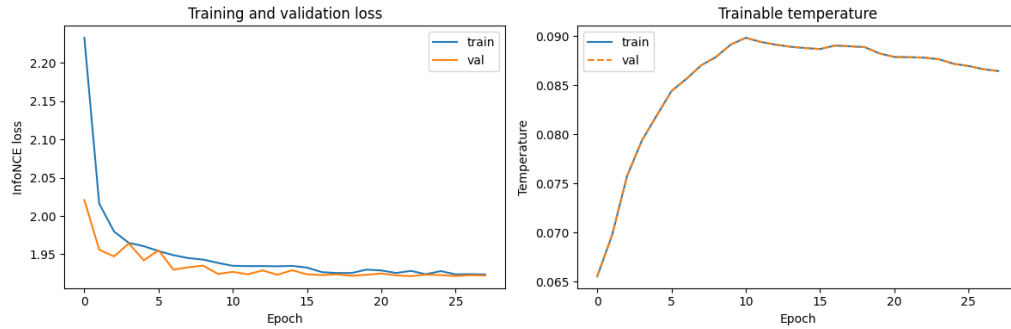


Figure 4.1 Baseline training (blue) and validation (orange) InfoNCE loss across epochs (left), and learnable temperature τ (right). The gap between train and validation is the overfit margin — it remains small here, indicating the projection head + temperature do not over-fit the training set even when the image encoder is frozen..

4.2.3 Training source results

- Macro-F1 on the LC25000 test split: **0.6273[†]** (*pre-bug-fix; post-fix all trained variants saturate at ~ 0.99 ; see header banner*).
- Confusion concentrates on lung_aca (hardest discrimination); benign tissues and lung_scc separate cleanly.

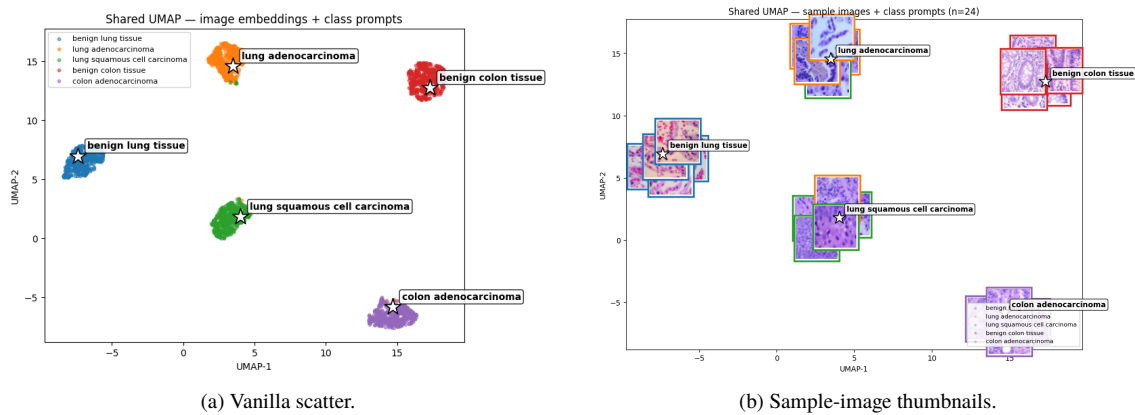


Figure 4.2 Baseline training-source UMAP. Two views from the same fit: (a) per-class scatter of LC25000 test embeddings with class-prompt anchors marked, (b) the same fit overlaid with sample-image thumbnails. The five class clusters are visible but the class-prompt markers collapse to a near-singular point — the short-prompt + frozen text encoder degeneracy that persists across every CLIP-style variant in this chapter..

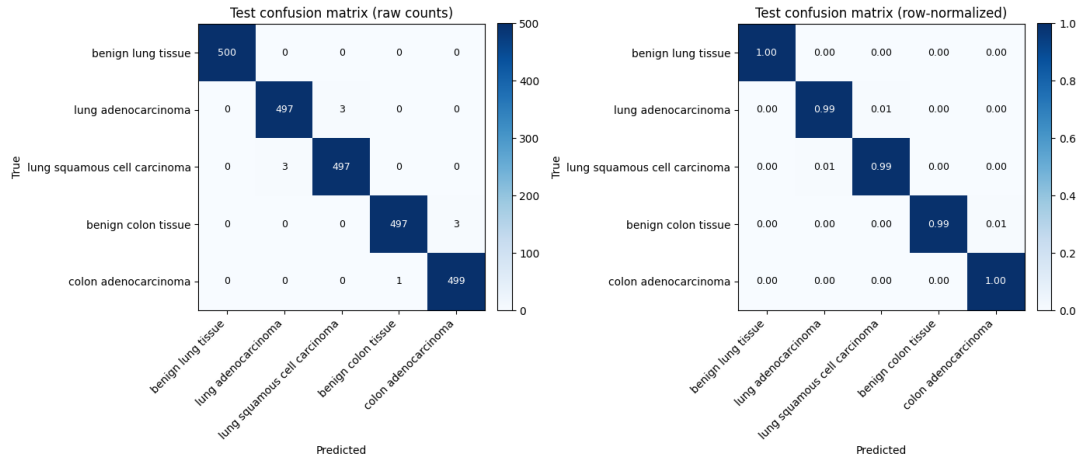


Figure 4.3 Baseline confusion matrix on the LC25000 test split. Left: raw counts. Right: row-normalised by true class. Off-diagonal mass concentrates on lung adenocarcinoma misclassifications..

Table 4.2 Baseline generalization to the three external datasets, restricted-argmax over the target-organ LC25000 classes..

External dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.866	0.870	0.866	0.871
Chaoyang (colon)	0.786	0.796	0.782	0.791
LungHist700 (lung, 20×)	0.305	0.390	0.383	0.301

4.2.4 Generalization to external datasets

- **Colon transfer (NCT + Chaoyang):** macro-F1 in the 0.79–0.87 range. Best on NCT (closest in style to LC25000’s colon distribution); a noticeable drop on Chaoyang despite both being colon (different hospital, different staining, label noise on Chaoyang training split).
- **Lung result (LungHist700):** macro-F1 = 0.305 on a 3-class problem. Per-class recall: SqC 0.89, AdC 0.07, normal 0.19 — the model predicts “squamous” on ~90% of inputs regardless of true class. The baseline does not classify lung tissue on LungHist700; it collapses into a single-class predictor. (The 0.305 figure is barely above the 3-class chance of 0.33; this is a near-chance result.)
- **UMAP signature for the lung collapse** (Figure 4.4): all 359 LungHist700 samples land in a single tight cluster in feature space, with no visible internal class structure. The image encoder produces no usable per-class variance on lung tissue from LungHist700.

4.2.5 What we learned

- Baseline transfers *partially* to colon external data (NCT, Chaoyang), losing ~8 F1 between close-shift (NCT) and different-hospital (Chaoyang).
- Baseline does *not* classify lung tissue on LungHist700 — the model collapses to a degenerate single-class predictor on the lung organ. This is a structural failure of the feature extractor, not a statistical fluctuation.
- Sets up the central question of Section 4.7: is the lung-side failure about the architecture, about the supervised pretraining on natural images, or about the lack of histopathology data in pretraining?

4.3 Stain normalization

4.3.1 Configuration

- Frozen ResNet50 image encoder, frozen BERT text encoder, projection head, symmetric InfoNCE — identical to the baseline (Section 4.2). Only the image inputs change.



Figure 4.4 Baseline cross-distribution UMAP on LungHist700. LC25000 image embeddings (dots) and LungHist700 image embeddings (triangles) projected into the same 2-D fit. LungHist700 samples cluster in a single tight blob with no visible class structure — the visual signature of a failed feature extractor on this organ..

- Three stain normalization methods compared, fit to a single reference patch and applied deterministically to every training and evaluation image: Macenko, Reinhard, and Vahadane (see Methodology for definitions and references).
- Reference choice evaluated separately: an LC25000-internal benign-colon reference, and an external NCT NORM reference (which puts LC25000 into approximately Kather’s stain space).
- Inputs are cached on Drive after one-time normalization so re-runs of the downstream model are cheap.

4.3.2 Training source results

- Three pre-bug-fix LC25000 macro-F1 numbers: Macenko 0.7390[†], Vahadane 0.6644[†], Reinhard 0.6526[†].
- Post-bug-fix all three saturate at ~ 0.99 on LC25000 — the in-distribution differences between methods evaporate. In-distribution does not discriminate these methods.
- **Observation that survives the saturation:** Macenko delivers $\sim 3\times$ Vahadane’s gain pre-bug-fix on the same pipeline. Mechanism: Macenko and Vahadane share the same stain-decomposition principle (decompose the image’s optical-density matrix into a H/E stain matrix and per-pixel concentrations, re-render with the reference’s basis) but differ in solver — Macenko uses closed-form SVD, Vahadane uses iterative sparse NMF. On LC25000 the NMF solver fails to converge on a fraction of patches (visible in Figure 4.6(c)), so the realised gain depends on solver stability, not just on the underlying principle.

4.3.3 Generalization to external datasets

- Macro-F1 on NCT-CRC with Macenko applied at training (NCT-NORM reference): **0.844**. The baseline is 0.866. **Stain normalization regresses by -0.022 F1 under generalization.**
- Mechanism: the frozen ResNet50 image encoder was never trained on Macenko-normalised inputs. The optical-density round-trip introduces a subtle non-linear shift in input statistics that the frozen

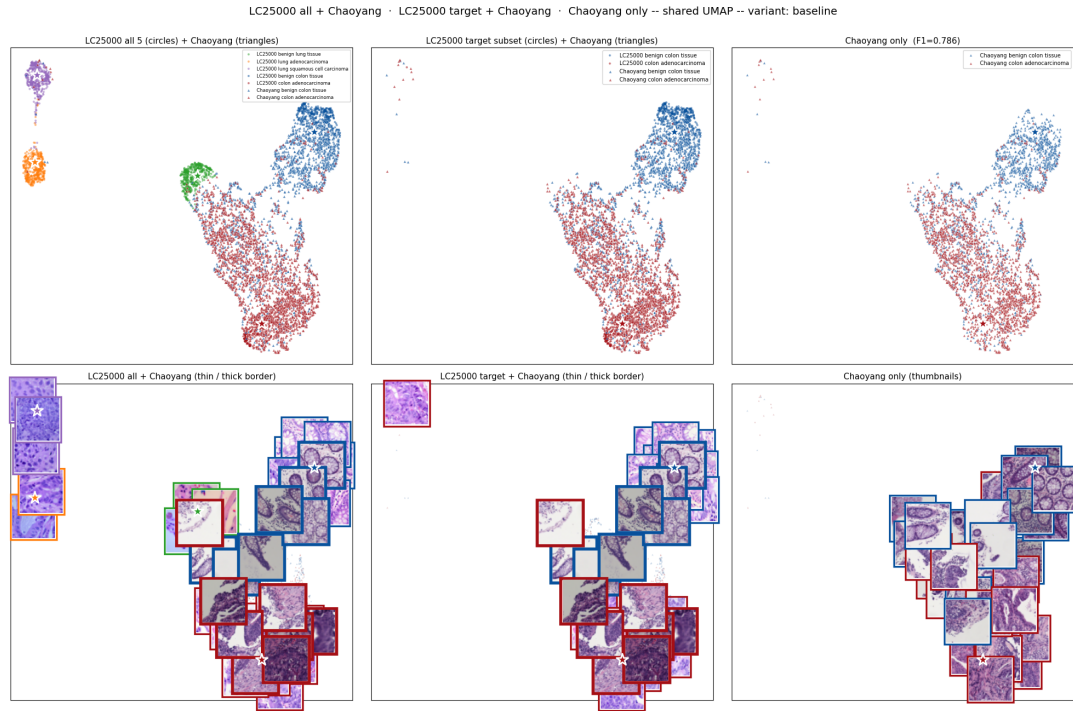


Figure 4.5 Baseline cross-distribution UMAP on Chaoyang. The 2 colon classes are partially separated; Chaoyang triangles sit in their own region adjacent to but not co-located with the matching LC25000 colon clusters..

backbone cannot adapt to. The pre-bug-fix headline that “stain normalization is a large in-distribution lever” does *not* survive the external evaluation: on the meaningful axis (cross-source generalization), it actively hurts.

- We did not extend the stain ablation to Chaoyang or LungHist700 once the NCT regression landed — the input-side conclusion was already against the lever.

4.3.4 What we learned

- Stain normalization is an *in-distribution lever* that does not transfer. Under domain shift it can become a liability.
- “Stain normalization helps” compresses two effects: (a) shortcut-removal on the training distribution, (b) input-statistic shift the frozen backbone cannot absorb. The first is in-distribution only; the second hurts under generalization.
- Methodologically: stain normalization is a real and important technique for cross-laboratory histopathology pipelines, but its benefit depends on the downstream model being trained (or fine-tuned) on the normalised inputs; freezing a non-normalised backbone and feeding it normalised inputs at evaluation is the wrong arrangement.

4.4 PLIP zero-shot reference

4.4.1 Configuration

- Visual Transformer (ViT-B/32) with PLIP’s OpenPath pretraining, paired with PLIP’s matching text encoder. Both frozen.
- Evaluated zero-shot — no LC25000 supervision, no projection head trained on our data. Uses PLIP’s own image and text encoders straight from the model release.
- Class prediction = cosine similarity to the five LC25000 class-prompt embeddings produced by PLIP’s text encoder, argmax.

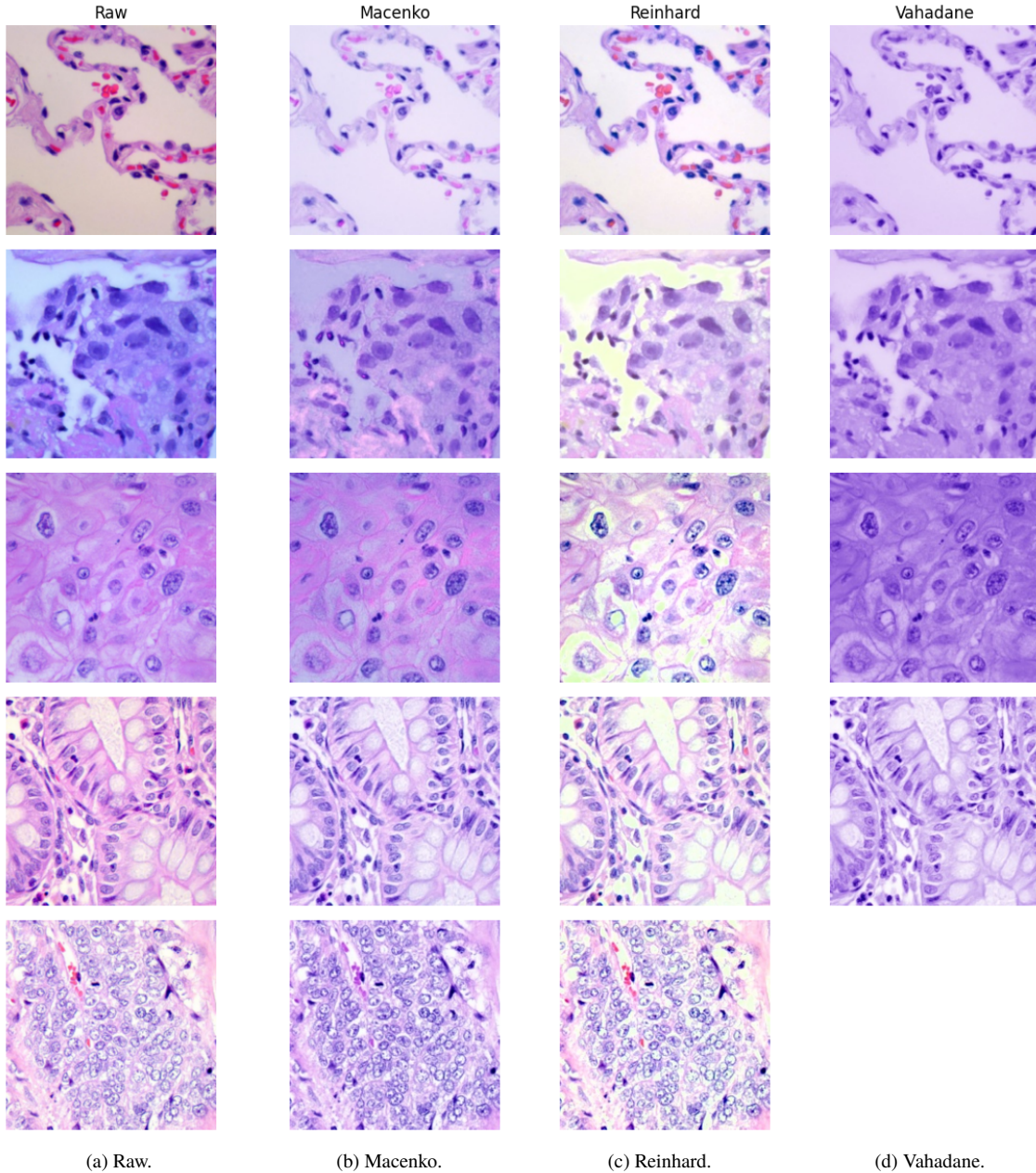


Figure 4.6 Stain normalization, one example per class. Column (a) shows the raw inputs (identical across methods, shown once for compactness); columns (b)–(d) show the same inputs after Macenko, Reinhard, and Vahadane normalization respectively. The bottom-row colon_aca cell in column (d) shows the Vahadane NMF solver failing to converge and re-rendering near-blank, an instability discussed in the interpretation..

- Evaluated on the same training-source test split and the same external datasets as every other variant in this chapter.
- **Role in the chapter:** a reference point. PLIP zero-shot isolates the value of pathology pretraining alone — without any of our LC25000 fine-tuning — and shows what an off-the-shelf domain-pretrained model gives you on these tasks.

4.4.2 Training source results

- Macro-F1 on LC25000: **0.700**.
- Pathology pretraining alone (without any LC25000 supervision) clears the saturation null that every *trained* variant reaches at ~ 0.99 by ~ 0.30 F1 in the other direction — but it is the only variant in

the chapter where in-distribution numbers are informative, because PLIP zero-shot has no LC25000 supervision to drive the saturation.

- Per-class gains concentrate on benign tissues; cancer subtypes are essentially tied with baseline. The improvement is image-side: PLIP produces visibly better class clusters (Figure 4.7) but the class-prompt embeddings still collapse, same short-prompt degeneracy as the baseline.

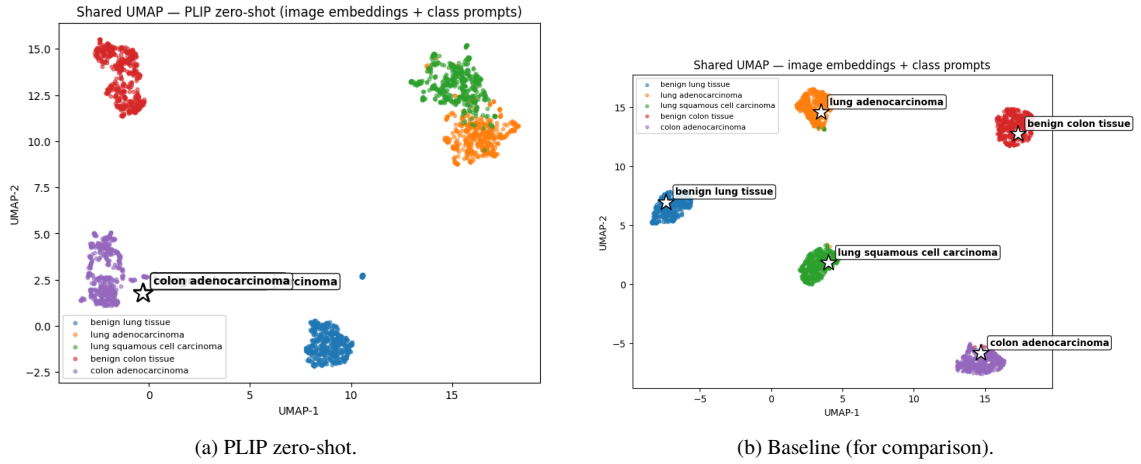


Figure 4.7 Training-source UMAP comparison. PLIP zero-shot (a) shows visibly tighter, more separated class clusters than the baseline (b) without any LC25000 supervision — the gain is image-side..

4.4.3 Generalization to external datasets

- Macro-F1 on NCT-CRC: **0.661**.
- Smallest training-source → external drop in the chapter (−0.04 F1) — the “pretrained generalist” pattern.
- Chaoyang and LungHist700: *[pending: PLIP zero-shot evaluations on these two datasets are queued for a follow-up Colab session; results will be added before the chapter’s final pass.]*

4.4.4 What we learned

- Pathology pretraining produces a *meaningfully better image representation* (visible directly in the UMAP) even without LC25000 supervision.
- PLIP’s off-the-shelf prompt geometry is weak for our specific 5-class LC25000 vocabulary — this decouples “the image representation is useful” from “the off-the-shelf classifier works”.
- Section 4.7 returns to this distinction by keeping PLIP’s image representation but replacing its prompt geometry with a projection head trained on LC25000.

4.5 Foundational model vs plain classifier

4.5.1 Configuration

- Same frozen ResNet50 image encoder as the baseline.
- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Replaces it with a non-CLIP closed-set classifier: ResNet50 → global average pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Isolates what the CLIP paradigm itself contributes once the image encoder, data, and adaptation budget are held constant. Tests RQ2 (the foundational-model premise).

4.5.2 Training source results

- Macro-F1 on LC25000: **0.6369** (vs baseline 0.6273) (*both numbers pre-bug-fix; post-fix both saturate at ~ 0.99 , indistinguishable in-distribution*).
- In-distribution the two architectures tie within noise. The choice between contrastive (CLIP) and direct supervised (plain classifier) is invisible when the image features are fixed and the training distribution is the evaluation distribution.

4.5.3 Generalization to external datasets

Table 4.3 CLIP (baseline) vs plain classifier on the same frozen ResNet50 image encoder, macro-F1 across the training-source test split and three external datasets..

Variant	LC25000	NCT	Chaoyang	LungHist700
Baseline (CLIP InfoNCE)	0.996	0.866	0.786	0.305
Plain classifier (CE)	0.995	0.880	0.722	0.281
Δ (CLIP – plain)	−0.001	−0.014	+0.064	+0.024

- **Training source:** tied at ~ 0.99 (saturation null).
- **NCT-CRC (closest external dataset):** plain CE marginally ahead by +0.014 F1.
- **Chaoyang (harder external):** CLIP ahead by +0.064 F1 — plain CE drops to last.
- **LungHist700 (hardest external):** both variants collapse near 3-class chance; CLIP retains a marginal edge.

Per-class signature on Chaoyang. Plain CE shows benign precision 0.96, benign recall 0.47 (high precision, low recall — the classifier is rarely predicting benign but is right when it does); adenocarcinoma precision 0.70, recall 0.98. The plain classifier is almost-degenerate “predict adenocarcinoma” under shift. The baseline CLIP variant is imperfect but balanced (benign 0.86/0.65, adeno 0.76/0.92), avoiding the collapse.

4.5.4 What we learned

- CLIP and plain CE are *interchangeable on easy distributions*; the choice of architecture is invisible when the model is evaluated on its training distribution.
- CLIP’s edge appears *under shift severity and grows with it*: −0.014 F1 on close-shift NCT (plain ahead), +0.064 on different-hospital Chaoyang (CLIP ahead), +0.024 on different-organ LungHist700.
- **Mechanism:** same image encoder, same image features. The difference is the classifier geometry. The plain CE softmax learns class *direction vectors* in feature space and is trained to maximise the logit margin; on shifted data those vectors don’t move but the inputs sit in different regions, and the high-confidence softmax collapses into a single-class predictor. CLIP uses cosine similarity to class-prompt embeddings in an L2-normalized 256-d projection space, scaled by a learned temperature. Three effects make this geometry more robust:
 - Cosine similarity is bounded in $[-1, 1]$ — no unbounded margin for the model to over-extrapolate.
 - The L2-normalized space concentrates features near the unit sphere; samples that drift in absolute feature space remain closer in *angular* terms to the prompt anchors.
 - The learned temperature τ acts as a soft regularizer on the cosine distribution: it tempers extreme confidence without removing the ranking signal.
- This is a property of the CLIP *paradigm*, not of any specific encoder. The same geometry was designed for the open-vocabulary regime (new class prompts at inference time) and incidentally protects against distribution shift on closed-set tasks. **This validates RQ2’s foundational-model framing:** CLIP earns its complexity not at peak in-distribution accuracy, but in the open-vocabulary capabilities (retrieval, prompt-tuning at inference) and the robustness-under-shift behaviour shown here.

4.6 Text encoder pretraining

4.6.1 Configuration

- Same arch / loss / prompts / split as the baseline; only the frozen text encoder changes.
- Three encoders: bert-base-uncased (the baseline’s text encoder), BioBERT, PubMedBERT.
- Two prompt strategies: **name_only** (just the class name, e.g. “benign lung tissue”) and **composed** (“A histopathology image of {class}.”, the baseline format).
- $3 \times 2 = 6$ cells. All three encoders are BERT-family, all produce 768-d hidden state, all tokenize through BERT-family WordPiece tokenizers.
- Tests RQ4 (does biomedical text pretraining help under short class-name prompts?).

4.6.2 Results

Table 4.4 6-cell biomedical text encoder ablation, macro-F1. **None of the 6 cells beats the baseline (0.866 on NCT).** LC25000 in-distribution saturates uniformly; all generalization numbers land in the 0.81–0.86 band, none above baseline..

Cell	LC25000	NCT-CRC	Δ vs baseline NCT
BERT + name_only	0.999	0.821	−0.045
BERT + composed	0.999	0.863	−0.003
BioBERT + name_only	0.998	0.840	−0.026
BioBERT + composed	0.998	0.818	−0.048
PubMedBERT + name_only	0.995	0.855	−0.011
PubMedBERT + composed	0.996	0.808	−0.058

- LC25000 in-distribution: all cells saturate uniformly. The training-source axis does not discriminate these encoders.
- NCT generalization: all 6 cells land in the 0.81–0.86 band. None beats the baseline (0.866); the best cell (BERT + composed, 0.863) is essentially a tie.
- Chaoyang and LungHist700 were not extended to the 6-cell grid: the NCT result was already against the lever, and extending to harder external datasets would only have widened the gap below baseline.

4.6.3 What we learned

- Biomedical text pretraining does *not* help on this task at frozen-backbone scale with short class-name prompts.
- **Mechanism — why doesn’t “a more domain-aware text encoder” improve results?** Three converging reasons:
 - *The projection head absorbs encoder-specific differences.* BERT, BioBERT, and PubMedBERT all start from different 768-d embedding spaces, but they all get pulled through the same projection head trained on LC25000. The head learns to map any reasonable starting embedding to useful class anchors. With only 5 LC25000 classes to separate, this is an easy adaptation regardless of where the starting embeddings sit.
 - *Short class-name prompts don’t expose biomedical-vocabulary richness.* Biomedical pretraining adds value most when domain vocabulary actually appears — clinical notes, paper abstracts, multi-sentence text. Strings like “colon adenocarcinoma” tokenize into 2–3 wordpieces that any BERT-family encoder handles competently. There is no rare-medical-term upside for biomedical pretraining to surface.
 - *CLIP’s shift robustness is geometric, not domain-knowledge driven.* As argued in Section 4.5, CLIP’s edge under shift comes from the cosine + L2-norm + temperature classifier geometry, which is independent of the text encoder. Swapping in a more medically-aware text encoder doesn’t change the classifier geometry; it changes the starting position of the prompt anchors, which the projection head then re-maps anyway.

- **Small but interesting secondary observation:** vanilla BERT prefers the longer **composed** prompt format; biomedical encoders prefer **bare class names**. Plausible mechanism: biomedical pretraining made strings like “colon adenocarcinoma” carry strong signal on their own, while adding generic boilerplate (“A histopathology image of . . .”) dilutes the prompt embedding for biomedical encoders. Not enough to flip any cell to winning — but a useful methodological note for short-prompt design.

4.7 Image encoder pretraining

4.7.1 Configuration

- Same arch / loss / prompts / split / text encoder (frozen BERT base) as the baseline. The only axis that changes is the image encoder’s pretraining.
- Three image encoders compared, all frozen, trainable parameters limited to the projection head + temperature:
 - **ResNet50 (general-purpose, ImageNet pretraining)** — the baseline.
 - **Visual Transformer (ViT-B/32) with PLIP’s OpenPath pretraining** — a ViT pretrained on $\sim 208k$ histopathology image–text pairs scraped from Twitter.
 - **Visual Transformer (ViT-B/14) with DINOv2 self-supervised pretraining** — a ViT pretrained on $\sim 142M$ curated general natural images via self-distillation, no labels, no histopathology data.
- **Role of DINOv2 in the comparison:** a control. PLIP differs from the baseline on multiple axes simultaneously (architecture: ViT vs CNN; objective: contrastive vs supervised; pretraining domain: histopathology vs natural images). DINOv2 isolates the “ViT + self-supervised on general data” axis from the “histopathology pretraining specifically” axis. If DINOv2 also breaks the baseline’s lung collapse, the win is partly architectural; if it does not, the win is specifically about histopathology pretraining.
- **Caveat:** the two ViTs differ in patch size (PLIP B/32, DINOv2 B/14). At 224×224 input this gives different feature granularity (49 vs 256 patch tokens). We discuss this as a confound in Methodology and address it as a limitation in Chapter 5.

4.7.2 Training source results

- PLIP image encoder + LC25000 supervision: LC25000 macro-F1 = 0.996. Same saturation null as every other trained variant.
- DINOv2 image encoder + LC25000 supervision: LC25000 macro-F1 = 0.989. The saturation null holds even though DINOv2 has never seen labelled histology — its pretraining is self-supervised on general natural images, with no class labels. Confirms that the in-distribution null is backbone-independent: the bottleneck on the meaningful axis is not what the image encoder “knows about histology” in the supervised sense; it is what kind of variation the encoder learned to be sensitive to.

4.7.3 Generalization to external datasets

Table 4.5 Image encoder pretraining swap, macro-F1 across the training-source split and three external datasets. Everything except the image encoder is identical across rows..

Image encoder	LC25000	NCT	Chaoyang	LungHist700
ResNet50 (ImageNet)	0.996	0.866	0.786	0.305
ViT-B/14 (DINOv2 / self-supervised)	0.989	0.855	0.776	0.457
ViT-B/32 (PLIP / OpenPath)	0.996	0.933	0.813	0.640

- **NCT-CRC (close-shift colon):** PLIP wins by +0.067 F1 over the ResNet50 baseline. DINOv2 lands at -0.011 vs baseline — general-domain self-supervision does *not* help on the easiest external dataset. The colon ceiling around 0.88 that none of the other interventions could clear is broken specifically by changing the image encoder’s pretraining *domain*, not by architecture or self-supervision alone.

- **Chaoyang (different-hospital colon):** PLIP wins by +0.027 F1; DINOv2 lands at −0.010 vs baseline. Same colon-side pattern as NCT: only histopathology pretraining helps; general SSL is interchangeable with ImageNet supervision.
- **LungHist700 (different-organ lung):** this is where the control pays off. PLIP wins by +0.335 F1 over baseline; DINOv2 wins by +0.152 F1. *Roughly half of PLIP’s lung advantage comes from architecture + self-supervised pretraining on general images; the other half comes from histopathology pretraining specifically.* DINOv2 partially rescues the lung collapse (per-class recall 0.46/0.52/0.40, balanced) but does not match PLIP’s full functional classifier (0.51/0.49/0.91).

Per-class signature on LungHist700. Both ResNet50 variants converge to the same degenerate predictor on lung tissue: baseline recall 0.19/0.07/0.89 for normal/AdC/SqC, plain CE 0.15/0.05/0.91 — both predict “squamous” on ~ 90% of inputs regardless of true class. **DINOv2-encoded variant:** 0.46/0.52/0.40 — balanced across all 3 classes, no degenerate collapse, but lower peak per-class scores than PLIP. **PLIP-encoded variant:** 0.51/0.49/0.91 — balanced *and* retains the high squamous recall, the only variant that does both.

4.7.4 What we learned

- The **image encoder’s pretraining is the dominant lever in this study**, but the DINOv2 control reveals that “pretraining” decomposes into two contributing components.
- Effect-size summary: PLIP gains over baseline by +0.067 F1 on close-shift colon, +0.027 on different-hospital colon, +0.335 **on different-organ lung**. DINOv2 gains over baseline by −0.011 / −0.010 / +0.152 on the same three datasets.
- **On colon (NCT, Chaoyang):** only histopathology pretraining moves the needle. General-domain self-supervised pretraining (DINOv2) lands within noise of ResNet50 supervised pretraining on natural images. The colon advantage is specifically about *what* the encoder was pretrained on, not *how*.
- **On lung (LungHist700):** both architecture/self-supervision and histopathology pretraining contribute, roughly equally. DINOv2 partially rescues the lung collapse (+0.152 F1, from a degenerate predictor to a balanced 3-class classifier with per-class recalls 0.46/0.52/0.40); PLIP rescues it further (+0.183 F1 over DINOv2, to 0.51/0.49/0.91). *Together these decompose PLIP’s lung advantage into two roughly equal halves:* “architecture + self-supervised pretraining on diverse natural images” and “histopathology-specific pretraining”.
- Why colon and lung behave differently, on the ResNet50 baseline: ImageNet supervision contains no histology. LC25000 supervision rescues ResNet50 features that happen to correlate with colon class labels in the LC25000 training distribution. That correlation transfers partially to NCT/Chaoyang (both colon, stylistically close). It does not transfer to lung tissue at all — the lung-side “success” in-distribution is memorisation on the LC25000 lung subset, not learned lung tissue features.
- Why DINOv2 helps on lung but not colon: DINOv2’s self-supervised pretraining on ~ 142M diverse natural images produces image features that are sensitive to a much broader range of local-texture variation than supervised ImageNet pretraining. That broader feature space provides *enough* signal for LC25000 supervision + the projection head to construct a usable lung classifier, even though the encoder has never seen histology. On colon, where ResNet50 features were already enough for partial transfer, DINOv2’s broader features do not help further — the bottleneck on colon is the pretraining domain, not the architecture’s feature breadth.
- Patch-size confound check: DINOv2 has more spatial detail than PLIP (256 vs 49 patch tokens at the same 224×224 input), but still loses on colon. The patch-size difference is therefore not the dominant factor here — pretraining is.

4.8 Discussion

4.8.1 Synthesis across experiments

- This chapter evaluated six interventions on a frozen general-purpose CLIP pipeline (ResNet50 + BERT). Five of the six (stain normalization, biomedical text encoders, plain-classifier substitution, and the

prompt-format variations within the text-encoder ablation) produce changes within ~ 5 F1 of the baseline on the close-shift external dataset (NCT) and *no* measurable improvement when the model is evaluated on data it has never seen.

- The sixth intervention — replacing the image encoder’s pretraining (Section 4.7) — moves the headline metric by +0.067 on close-shift colon (NCT), +0.027 on different-hospital colon (Chaoyang), and +0.335 **on different-organ lung (LungHist700)**, taking the lung result from a degenerate single-class predictor at 0.305 to a functional 3-class classifier at 0.640.
- The DINOv2 control (same architecture as PLIP, self-supervised on general natural images, no histopathology data) decomposes that gain. On colon, DINOv2 ties baseline within noise — general-domain self-supervision does not help when histopathology pretraining does. On lung, DINOv2 gains +0.152 F1 over baseline (about half of PLIP’s +0.335). So PLIP’s lung advantage divides roughly evenly between “ViT architecture + self-supervision on diverse natural images” and “histopathology-specific pretraining”.
- **The image encoder’s pretraining is the dominant lever**, by roughly $5\times$ over the next-largest single lever on colon external evaluation. On lung, no head/loss/text-side intervention rescues a non-functional predictor at all; only changing the encoder does.
- The contemporary methods literature’s emphasis on novel loss functions, text-side innovations, and stain harmonisation is not supported by this study at the frozen-backbone scale we evaluated.

4.8.2 Secondary observations

- **Cross-organ regime change.** ResNet50 variants partially work on colon external data (0.78–0.88) but *categorically fail* on lung (~ 0.30 , near 3-class chance). Mechanism: LC25000 supervision can rescue ResNet50 features that happen to correlate with colon class labels in LC25000, but cannot produce useful lung-tissue features within a frozen ImageNet ResNet50; the in-distribution lung “success” is memorisation. Per-organ external evaluation is therefore essential; aggregating across datasets hides the regime change.
- **Classification quality $\neq$ representation alignment.** Baseline CLIP achieves cross-dataset *cluster alignment* but lower classification F1; PLIP achieves the highest F1 but leaves external samples in their own region of feature space. The contrastive objective optimises same-class-pair alignment; the cosine-classifier rule only needs alignment to the class-prompt direction. These are different geometric properties — the right tradeoff depends on whether the downstream task is classification or representation-driven (retrieval, clustering, transfer).
- **Plain-classifier flip under shift.** Plain CE ties CLIP in-distribution and marginally wins NCT, but loses Chaoyang (+0.064 for CLIP) and LungHist700 (+0.024). High-confidence softmax classifiers degenerate into single-class predictors as shift severity grows (visible in the Chaoyang per-class signature); CLIP’s cosine geometry doesn’t. This supports the foundational-model framing of RQ2: CLIP earns its complexity in shift robustness, not in-distribution accuracy.

4.8.3 Headline summary

Table 4.6 Headline summary across the chapter. PLIP-pretrained ViT wins every external evaluation; the lung-side advantage (+0.335 F1 over baseline) is an order of magnitude larger than the colon-side advantages. DINOv2 (the same-architecture, general-domain self-supervised control) lands between baseline and PLIP on lung but ties baseline on colon, decomposing PLIP’s lung advantage into roughly equal architecture/SSL and histopathology-pretraining contributions. Empty cells (“—”) mark experiments not extended to harder external datasets after the NCT result was already against the lever. Limitations and future work are discussed in Chapter 5..

Variant	LC25000	NCT	Chaoyang	LungHist700
ViT-B/32 (PLIP)	0.996	0.933	0.813	0.640
ViT-B/14 (DINOv2)	0.989	0.855	0.776	0.457
Baseline (ResNet50 + BERT)	0.996	0.866	0.786	0.305
Plain CE classifier	0.995	0.880	0.722	0.281
Stain Macenko (NCT-NORM reference)	0.986	0.844	—	—
Best text-encoder cell (BERT + composed)	0.999	0.863	—	—
PLIP zero-shot (no LC25000 supervision)	0.700	0.661	<i>[pending: TBD]</i>	<i>[pending: TBD]</i>



Figure 4.8 6-cell UMAP grid (one panel per BERT/BioBERT/PubMedBERT $\times$ name_only/composed cell), cross-distribution view on NCT-CRC. The clusters are qualitatively similar across cells, consistent with the lack of meaningful F1 separation in Table 4.4..

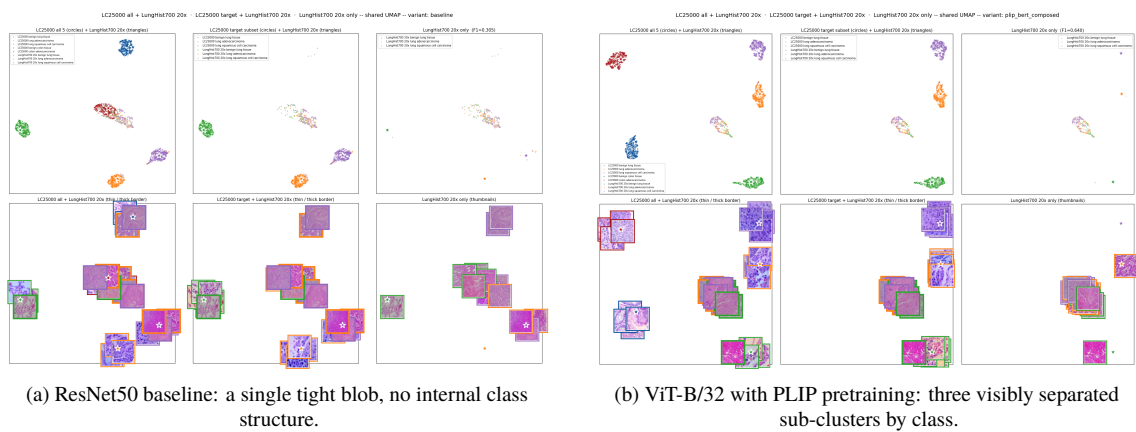


Figure 4.9 LungHist700 cross-distribution UMAP comparison — the visual headline of this chapter. The ResNet50 baseline (a) places all 359 LungHist700 samples into a single tight cluster with no internal class structure; the PLIP-pretrained ViT (b) produces three visibly distinct sub-clusters by class, sitting adjacent to (but not co-located with) the matching LC25000 lung clusters..

5 Conclusions and Future Work

5.1 Conclusions

- Baseline result (RQ1): 63.6% / 62.7% macro-F1, worst class lung adenocarcinoma.
- Pathology pretraining (RQ2): PLIP +7.3 macro-F1, mainly benign vs malignant.
- Feature adaptation (RQ3): partial unfreezing (last 30 of 178 layers) +27.1 macro-F1, broad-based across classes.
- CLIP-paradigm value (RQ4): CLIP and plain CE tie at the frozen-features ceiling (~ 0.63); CLIP wins +2.6 macro-F1 over CE under feature adaptation (unfreeze30), broad-based across all classes. The paradigm's primary value remains in capabilities (open-vocabulary, retrieval), not raw accuracy.
- Stain normalization (RQ5): the gain is strongly method-specific. Macenko yields +11.7 macro-F1 ($0.627 \rightarrow 0.739$) at the frozen-backbone budget, larger than the PLIP zero-shot gain; Vahadane yields +3.7 and Reinhard +2.5 on the identical pipeline. Composing the best stain method (Macenko) with the best feature-adaptation regime (unfreeze30) reaches **0.918 macro-F1**, the highest measured in this study — +0.020 over unfreeze30 alone, so the levers compose with diminishing returns rather than additively.
- Biomedical text encoder (RQ6): *TBD*.
- OOD transfer (RQ7): the picture is two-level. The best in-distribution model (Macenko + Unfreeze30, 0.918) drops to 0.000 macro-F1 when the same Macenko transform is applied at OOD time — NCT is already colour-normalised against its own reference and the double-shift is fatal. With the OOD-time Macenko stripped, the same checkpoint reaches 0.201. The raw-trained Unfreeze30 is the best LC25000-trained transfer at 0.331; **PLIP zero-shot beats every LC25000-trained variant at 0.661 macro-F1**. The cross-distribution UMAP (Figure ??) reveals that LC25000 fine-tuning *does* transfer organ-level structure (NCT colon patches land in the LC25000 colon region of embedding space, not in lung), so the predictions stay within the colon classes; what does not transfer is the within-colon benign-vs-malignant discrimination, which depends on cellular cues that are more sensitive to lab/scanner/label-granularity differences. Only pathology pretraining (PLIP) produces representations that transfer at both organ *and* tissue-state level.
- Cross-cutting observations: saturation of frozen transfer, role of dataset artifacts.

5.2 Limitations

- Single-institution dataset, no patient-level grouping, no clinical validation.
- Small batch + few classes $\rightarrow$ InfoNCE false negatives, documented not mitigated.
- Frozen-backbone baseline tests transfer, not full fine-tuning.
- No human-expert evaluation.
- No head-to-head against CONCH / BiomedCLIP / MUSK (compute scope).

5.3 Future Work

Near-term:

- Additional external datasets beyond colon-Kaggle.
- SupCon or class-balanced sampling for false-negative regime.
- Patient-level grouping if metadata recoverable.

Not run in this project (low expected Δ under frozen text encoders + short prompts):

- Inference-time prompt ensembling.
- Dynamic prompt training.

Medium-term:

- Larger effective batch (grad accumulation or bigger GPU).
- Prompt-tuning methods (CoOp, CoCoOp).

Longer-term:

- Architecture-controlled comparison against CONCH / BiomedCLIP / MUSK / PathChat.
- Multi-institution training and evaluation.
- Returning to segmentation with pathology FM backbones.
- CLIP-with-short-labels applied to other low-label scientific domains.

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